CLINICAL TRIAL FAILURES · DISCONTINUATION RATE · ONCOLOGY PHASE 2,3 · STARTS 2015–2024

# VEGF / VEGFR: 13 molecules stopped early

**20 stopped trials across 13 molecules: Cabozantinib, Lenvatinib, Bevacizumab, Catequentinib and 9 more.** Against 5.2% for all oncology trials with a resolved drug target and 4.5% across all oncology Phase 2,3 trials in the same window. The rate counts trials that stopped early for efficacy, safety or benefit–risk reasons. Trials that ran to completion and missed their primary endpoint are counted separately below, never in the rate.

**3.0%**
20 of 657 closed trials (95% CI 2.0%–4.7%)

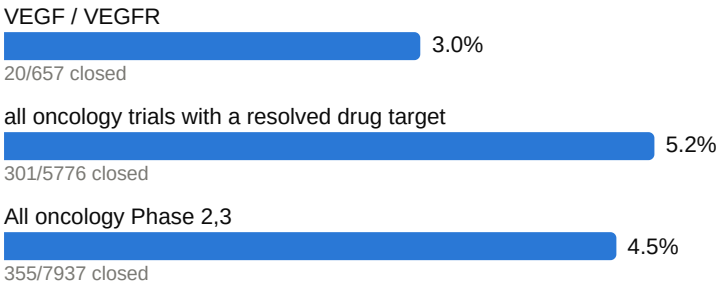
**13 / 5 / 0**
efficacy / safety / both (adds to 20)

**30.4%**
of 2161 trials have closed · 1504 still open or unresolved

**0.8%**
stopped for a biological reason within 3 years of starting (95% CI 0.5%–1.6%; 776 still at risk; 1.5% area-wide)

**14 of 28** **Completed and missed the primary endpoint.** Of 492 completed trials in this segment, 28 posted a primary result on ClinicalTrials.gov that can be read — the sponsor's own comparison held to the sponsor's own threshold, or its statement that the endpoint was missed: 14 missed and 10 met, 4 split across co-primary endpoints. Counted separately and never in the rate above; a floor, not a rate. 249 of the 459 completed trials that finished more than a year ago have posted no results at all — a gap in what can be known, not a sign of failure.

## How this compares




**What is in this class:** 470 distinct experimental drugs across 2161 trials (small molecule 1839, antibody 1278, unknown 360, protein 73). Most tested: BEVACIZUMAB (467), RIVOCERANIB (341), LENVATINIB (267), CATEQUENTINIB (259), PEMBROLIZUMAB (160). A class groups drugs by what they act on; check that the grouping is one you would make before reading the rate as a property of the mechanism.

**How much does this rest on one decision?** The 20 stops came from 20 sponsor-asset programmes across 18 sponsors; the largest (Yale University) contributed 1; removing that programme's trials from both sides leaves 2.9%. Screened alongside 85 other segments, the rate is within what screening that many classes could produce (q=1); the molecules and trials that failed here are the record either way. A further 38 closed trials here were terminated with no cause recorded in the registry; if every one of them were biological the rate would be 8.8%. Read it as a screen worth checking against the underlying trials.

Bars show the share of closed trials stopped for biological reasons. Confidence intervals overlap where sample sizes are small — read the counts, not just the bars.

## By start cohort

| Trials started | Stops | Closed | Rate | 95% CI |
|---|---|---|---|---|
| 2015–2017 | 7 | 240 | 2.9% | 1.4%–5.9% |
| 2018–2020 | 10 | 251 | 4.0% | 2.2%–7.2% |
| 2021–2024 | 3 | 166 | 1.8% | 0.6%–5.2% |

## Where the stops sit

Efficacy / futility: 13, Safety: 5, Benefit–risk: 2.
Sponsors with most stops: Jiangsu Alphamab Biopharmaceuticals Co., Ltd (2), Sun Yat-sen University (2), Yale University (1), Great Novel Therapeutics Biotech & Medicals Corporation (1), Niguarda Hospital (1), Eli Lilly (1).

## The stopped trials

| Trial | Ph | Sponsor | Experimental drugs | Stopped | Type | Registry stop reason |
|---|---|---|---|---|---|---|
| NCT04955743 | 2 | Yale University | PEMBROLIZUMAB, LENVATINIB | 2025-05 | Efficacy | Lack of activity in the drugs |

| Trial | Ph | Sponsor | Experimental drugs | Stopped | Type | Registry stop reason |
|---|---|---|---|---|---|---|
| NCT05770882 | 2 | Great Novel Therapeutics Biotech & Medicals Corporation | TUCIDINOSTAT, REGORAFENIB | 2025-01 | Benefit–risk | The overall profile does not support development for Hepatocellular Carcinoma |
| NCT03686345 | 2 | Niguarda Hospital | MIDOSTAURIN | 2023-06 | Efficacy | didn't reach the primary endpoint |
| NCT04145700 | 2 | Eli Lilly LLY | RAMUCIRUMAB | 2023-02 | Efficacy | Study terminated for meeting protocol specified futility criteria. |
| NCT05001724 | 3 | Jiangsu Alphamab Biopharmaceuticals Co., Ltd | ERFONRILIMAB, LENVATINIB | 2023-02 | Efficacy | Because the efficacy data did not meet expectations, the sponsor decided to terminate the study. |
| NCT04438902 | 2 | First Affiliated Hospital of Zhejiang University | OSIMERTINIB, CATEQUENTINIB | 2022-12 | Safety | Many patients can't tolerate the combination treatment due to adverse events. |

The 6 most recent of 20 stopped trials. Every molecule behind all 20 is named above; the complete trial list, the trials still running and the terminations with no readable cause ship with the dataset and the diligence report.

**Method**

Denominator: ClinicalTrials.gov interventional Phase 2,3 oncology trials started 2015–2024 that have closed (completed or terminated). Numerator: terminated trials whose registry stop reason is classified as biological (efficacy, safety or benefit–risk) by Classification V2 — held-out precision 95.5%, recall 95.3% (n=600). Drugs are linked to ChEMBL and the NCI Thesaurus; 76% of industry oncology trials in this window carry a resolved drug target, and a trial without one cannot enter a mechanism class. Intervals are Wilson 95%.

**Limits**

Not a failure rate: completed trials that missed their endpoints are not in the rate. They are counted separately where the sponsor posted a primary result that can be read — most did not, so that count is a floor — and programmes discontinued after a completed trial do not appear. Stop reasons are sponsor-reported. This is a closed-trial proportion, not a time-to-event analysis: only 30.4% of trials in this segment have closed, and a trial that stops early enters the denominator sooner than one that runs to completion, which can inflate the rate in immature segments. Recent cohorts have fewer closed trials, so their rates are less stable. Research signals, not clinical or investment advice.

**The whole cohort, trial by trial.** The diligence report names every trial behind these figures — the ones still running and the ones that completed and missed their primary endpoint included — and places your molecule against each one that failed. €99, delivered immediately: **clinicaltrialfailures.com/asset-check**

Sources: ClinicalTrials.gov (NLM); ChEMBL (EMBL-EBI, CC BY-SA 3.0); NCI Thesaurus (NCI); RxNorm/RxClass (NLM); SEC EDGAR. Classification, linkage and rates by Clinical Trial Failures. Rebuilt weekly; this brief covers trials started 2015–2024.